Problem 10

Prove the general formula for powers of cosine:

$$\cos^n x = 2^{1-n}\left[\cos nx + n\cos(n-2)x + \frac{n(n-1)}{1\cdot 2}\cos(n-4)x + \cdots + A\right]$$

where

$$A = \begin{cases} \frac{1}{2}\cdot\frac{n(n-1)\cdots(\frac{n}{2}+1)}{(\frac{n}{2})!} = \frac{1}{2}\binom{n}{n/2} & \text{if } n = 2k \text{ (even)} \\ \frac{n(n-1)\cdots(\frac{n+1}{2}+1)}{(\frac{n-1}{2})!} = \binom{n}{(n-1)/2} & \text{if } n = 2k+1 \text{ (odd)} \end{cases}$$

Solution

We will use Euler's formula and the binomial theorem to derive this general formula.

Setup Using Euler's Formula

From Euler's formula:

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

Therefore:

$$\cos^n x = \left(\frac{e^{ix} + e^{-ix}}{2}\right)^n = \frac{(e^{ix} + e^{-ix})^n}{2^n}$$

Binomial Expansion

Using the binomial theorem:

$$(e^{ix} + e^{-ix})^n = \sum_{k=0}^{n}\binom{n}{k}(e^{ix})^{n-k}(e^{-ix})^k = \sum_{k=0}^{n}\binom{n}{k}e^{i(n-2k)x}$$

Pairing Symmetric Terms

For any k and $n-k$, we have:

$$e^{i(n-2k)x} + e^{-i(n-2k)x} = 2\cos((n-2k)x)$$

We need to consider even and odd cases separately.

Case 1: $n = 2m$ (Even)

Step 1: Pair Terms

For even $n = 2m$, we can pair all terms except the middle term where $k = m$:

$$(e^{ix} + e^{-ix})^{2m} = 2\sum_{k=0}^{m-1}\binom{2m}{k}\cos((2m-2k)x) + \binom{2m}{m}$$

The middle term $\binom{2m}{m}$ corresponds to $k = m$, giving $e^{i\cdot 0} = 1$.

Step 2: Simplify

$$\begin{aligned}
\cos^{2m} x &= \frac{1}{2^{2m}} \left[2 \sum_{k=0}^{m-1} \binom{2m}{k} \cos((2m-2k)x) + \binom{2m}{m} \right] \\
&= \frac{1}{2^{2m-1}} \left[\sum_{k=0}^{m-1} \binom{2m}{k} \cos((2m-2k)x) + \frac{1}{2} \binom{2m}{m} \right] \\
&= 2^{1-2m} \left[\cos 2mx + \binom{2m}{1} \cos(2m-2)x + \binom{2m}{2} \cos(2m-4)x + \cdots + \frac{1}{2} \binom{2m}{m} \right]
\end{aligned}$$

Since $\binom{2m}{1} = 2m$, $\binom{2m}{2} = \frac{2m(2m-1)}{2}$, etc., we get:

$$\boxed{\cos^{2m} x = 2^{1-2m} \left[\cos 2mx + 2m \cos(2m-2)x + \frac{2m(2m-1)}{2} \cos(2m-4)x + \cdots + \frac{1}{2} \binom{2m}{m} \right]}$$

where $A = \frac{1}{2}\binom{2m}{m} = \frac{1}{2} \cdot \frac{2m!}{m! \cdot m!}$.

Case 2: $n = 2m+1$ (Odd)

Step 1: Pair Terms

For odd $n = 2m+1$, all terms can be paired (no middle term):

$$(e^{ix} + e^{-ix})^{2m+1} = 2 \sum_{k=0}^{m} \binom{2m+1}{k} \cos((2m+1-2k)x)$$

Step 2: Simplify

$$\begin{aligned}
\cos^{2m+1} x &= \frac{2}{2^{2m+1}} \sum_{k=0}^{m} \binom{2m+1}{k} \cos((2m+1-2k)x) \\
&= \frac{1}{2^{2m}} \sum_{k=0}^{m} \binom{2m+1}{k} \cos((2m+1-2k)x) \\
&= 2^{-2m} \left[\cos(2m+1)x + (2m+1) \cos(2m-1)x + \cdots + \binom{2m+1}{m} \cos x \right]
\end{aligned}$$

$$\boxed{\cos^{2m+1} x = 2^{-2m} \left[\cos(2m+1)x + (2m+1) \cos(2m-1)x + \cdots + \binom{2m+1}{m} \right]}$$

where $A = \binom{2m+1}{m} = \frac{(2m+1)!}{m! \cdot (m+1)!}$.

General Formula

Combining both cases, we have for general n:

$$\cos^n x = 2^{1-n} \left[\cos nx + n \cos(n-2)x + \frac{n(n-1)}{2} \cos(n-4)x + \cdots + A \right]$$

where:

- If n is even: $A = \frac{1}{2}\binom{n}{n/2}$
- If n is odd: $A = \binom{n}{(n-1)/2}$

Verification: Special Cases

For $n = 2$

$$\cos^2 x = 2^{-1}\left[\cos 2x + 2\cos 0 \cdot \frac{1}{2}\right] = \frac{1}{2}[\cos 2x + 1]$$

This matches the known formula $\cos^2 x = \frac{1+\cos 2x}{2}$.

For $n = 3$

$$\cos^3 x = 2^{-2}\left[\cos 3x + 3\cos x\right] = \frac{1}{4}[\cos 3x + 3\cos x]$$

This matches the known formula.

For $n = 4$

$$\cos^4 x = 2^{-3}\left[\cos 4x + 4\cos 2x + \frac{4\cdot 3}{2} \cdot \frac{1}{2}\right] = \frac{1}{8}[\cos 4x + 4\cos 2x + 3]$$

This can be verified using the identity $\cos^4 x = \frac{3+4\cos 2x+\cos 4x}{8}$.

Final Answer

$$\boxed{\cos^n x = 2^{1-n}\left[\sum_{k=0}^{\lfloor n/2\rfloor} \binom{n}{k} \cos((n-2k)x) \cdot \delta_k\right]}$$

where $\delta_k = 1$ for $k < n/2$ and $\delta_{n/2} = \frac{1}{2}$ when n is even.
Explicitly:

$$\boxed{\cos^n x = 2^{1-n}\left[\cos nx + n\cos(n-2)x + \frac{n(n-1)}{2}\cos(n-4)x + \cdots + A\right]}$$

with:

$$A = \begin{cases} \frac{1}{2}\binom{n}{n/2} & \text{if } n \text{ is even} \\ \binom{n}{(n-1)/2} & \text{if } n \text{ is odd} \end{cases}$$

This formula expresses any integer power of cosine as a linear combination of cosines of multiple angles.